

Weather Field Journal

Weeks 5–6 • Chapter 21 Weather

This workbook turns your weather journal into an interactive digital notebook. You can type into the boxes, check answers, sort ideas, compare weather systems, decode station models, and work through the chapter review without a separate sheet of paper.

How to use this workbook

- 1. Type in the fill-in-the-blank and short-answer boxes.
- 2. Use the checkboxes and quick sort sections to interact with the lesson.
- 3. Keep answers brief. This is a guided journal, not a full rewrite of the textbook.
- 4. If you want, you can still sketch on a printed copy or on a tablet with annotation tools.

Student Name:

Class/Section:

Date:

Confidence before we start (1–5):

A. Word Parts Builder

Use the chapter vocabulary to complete the table. The chapter itself highlights word parts like iso- and therm in isotherm and uses a word-parts table as a reading tool.

Word	What it means in weather	Root	Prefix	Suffix
isobar				
isotherm				
forecast				
meteorology				

B. Two-Column Notes

The textbook recommends two-column notes for Sections 1 and 4. Instead of blank notebook paper, use these guided rows for key ideas and details.

Key term, italicized word, or main idea	Definition, details, example, or why it matters

Section 1 • Air Masses

1. Know It

Complete the key ideas.

An air mass is a large body of air in which

temperature and moisture are

2. Sort It

Match the source-region clue to the type of air mass.

forms over land

→ symbol/name:

Usually brings

forms over oceans

→ symbol/name:

Usually brings

forms in tropical regions

→ symbol/name:

Usually brings

forms in polar regions

→ symbol/name:

Usually brings

3. Use It

Forecast Desk: If the air in your area is warm and dry, which air mass is most likely responsible? Write the type and the letters used to designate it.

Type of air mass:

Letters used to designate it:

Section 2 • Fronts

1. Compare Cold, Warm, Stationary, and Occluded Fronts

Front	How it forms	Typical weather	Sketch/keyword
Cold front			
Warm front			
Stationary front			
Occluded front			

2. Quick Check

Which type of front forms when two air masses move parallel to the boundary between them?

Section 2 • Midlatitude Cyclones + Severe Weather

1. Sequence the Stages of a Midlatitude Cyclone

Stage 1	Stage 2	Stage 3	Stage 4
Stage name: 	Stage name: 	Stage name: 	Stage name:
What happens? 	What happens? 	What happens? 	What happens?

2. Severe Weather Snapshot

Thunderstorm	A usually brief, heavy storm with rain, strong winds, lightning, and thunder.
Hurricane	A severe tropical storm with winds more than 120 km/h spiraling toward a low-pressure center.
Lightning	A giant spark that equalizes electrical charges in or between clouds and the ground. Write one memory h

Section 3–4 • Instruments + Weather Maps

1. Instrument Match-Up

thermometer



barometer



anemometer



wind vane



radiosonde



radar



2. Decode a Station Model

The textbook explains that a station model can show cloud cover, wind speed, wind direction, weather conditions, temperature, dew point, pressure, and pressure trend. Use the prompts below to practice reading one.



Temperature:

Dew point:

Wind direction:

Wind speed:

Pressure:

What kind of weather might you expect?

Section 4 • Forecasting the Weather

1. Forecast Types

Nowcast Focuses on:

Daily forecast Looks ahead:

Extended / medium-range Looks ahead:

Long-range forecast Looks ahead:

2. Watch or Warning?

A watch means conditions are favorable. A warning means severe weather has been spotted or is expected soon. Decide which one belongs in each situation.

1. Meteorologists see atmospheric conditions that could produce tornadoes later today.

2. A tornado has been spotted near your county.

3. A hurricane is expected within 24 hours.

3. Safety Snapshot

Choose one: During a thunderstorm, I should avoid running water and electrical appliances.

☐ True ☐ False

Chapter 21 Review Challenge

1. Use the Term Correctly

air mass

stationary front

station model

2. Explain the Difference

midlatitude cyclone and hurricane

wind vane and anemometer

radiosonde and radar

3. Quick Multiple Choice

The eye of a hurricane is a region of:

Chapter 21 Review Challenge

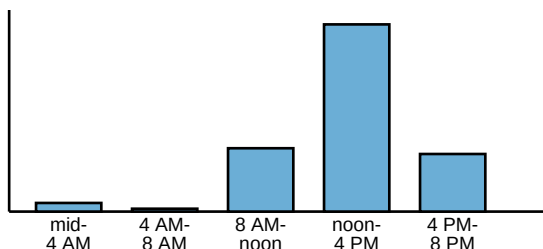
4. Short Answer

1. What causes lightning?

2. What is the most likely location for hurricane development? Explain why.

3. How could a meteorologist use a station model to determine whether a cold front is approaching?

5. Interpret the Tornado Graph



Least tornadoes occur at:

Most tornadoes occur at:

Would students be more likely to experience a tornado at school or at home?

Wrap-Up

6. My Weather Learning Reflection

The idea I understand best now is...

The weather topic I still need to review is...

One activity in this workbook that helped me learn was...

If I were giving a weather forecast tomorrow, I would pay closest attention to...

Teacher / Parent Check-In

Strengths I noticed:

Next step or review suggestion: